

T7.2.3-Worksheet 1-Structure Diagram

Answer the following questions.

1. Draw a flow that

- Inputs a number
- Find out number is even or odd (using MOD function)
- Output Even or Odd

Answer based on the discussion in class

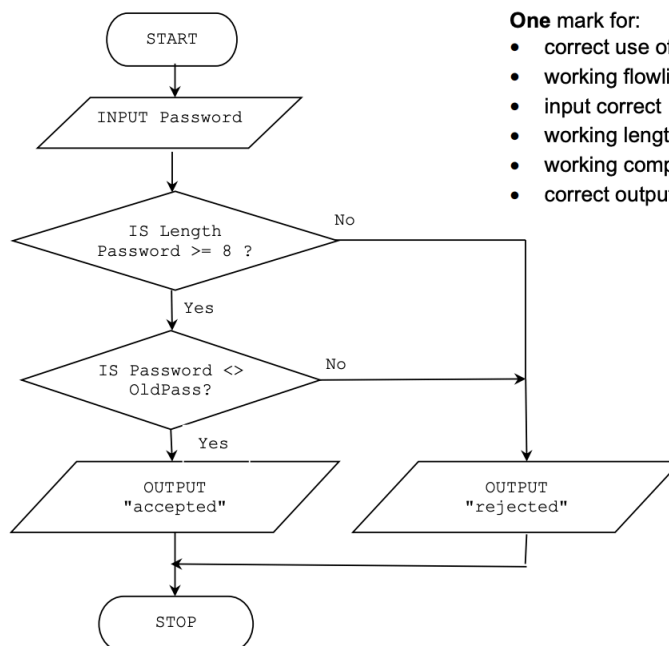
2.

A programmer is designing a program to check the length of a password and to check password input is the same as the stored password.

The program requirements are:

- input the password, `Password`
- check if there are at least 8 characters in the password
- check that the password is **not** the same as the stored password `OldPass`
- output 'accepted' if both tests are completed successfully
- otherwise, output 'rejected'.

Use the variable names given.



One mark for:

- correct use of flowchart symbols
- working flowlines
- input correct
- working length check
- working comparison
- correct output messages

Answer:

3. An algorithm needs to total 50 numbers between 1 and 100 inclusive.

Draw a flowchart that:

- Uses a count-controlled loop from 1 to 50
- Uses an appropriate prompt to ask for a number between 1 and 100
- Totals the numbers as they are entered
- Outputs the total after the loop has completed with an appropriate message.

Answer:

One mark per mark point

- Two labelled terminators START and STOP
- Loop initialisation, incrementation and termination
- Input with prompt
- Total initialised and totalling of inputs
- Output with message
- Fully correct flowchart with all correct arrows, yes/no labels and symbols

